22. For balance, let me note that there are physicists and philosophers who do not agree with this conclusion. Even though Einstein gave up on Mach's principle, during the last thirty years it has taken on a life of its own. Various versions and interpretations of Mach's idea have been put forward, and, for example, some physicists have suggested that general relativity *does* fundamentally embrace Mach's ideas; it's just that some particular shapes that spacetime can have—such as the infinite flat spacetime of an empty universe—don't. Perhaps, they suggest, any spacetime that is remotely realistic—populated by stars and galaxies, and so forth—does satisfy Mach's principle. Others have offered reformulations of Mach's principle in which the issue is no longer how objects, such as rocks tied by a string or buckets filled with water, behave in an otherwise empty universe, but rather how the various time slicings—the various three-dimensional spatial geometries—relate to one another through time. An enlightening reference on modern thinking about these ideas is *Mach's Principle: From Newton's Bucket to Quantum Gravity,* Julian Barbour and Herbert Pfister, eds. (Berlin: Birkhäuser, 1995), which is a collection of essays on the subject. As an interesting aside, this reference contains a poll of roughly forty physicists and philosophers regarding their view on Mach's principle. Most (more than 90 percent) agreed that general relativity does not fully conform to Mach's ideas. Another excellent and extremely interesting discussion of these ideas, from a distinctly pro-Machian perspective and at a level suited to general readers, is Julian Barbour's book *The End of Time: The Next Revolution in Physics* (Oxford: Oxford University Press, 1999).

23. The mathematically inclined reader might find it enlightening to learn that Einstein believed that spacetime had no existence independent of its metric (the mathematical device that gives distance relations in spacetime), so that if one were to remove everything—including the metric—spacetime would *not* be a something. By "spacetime" I always mean a manifold together with a metric that solves the Einstein equations, and so the conclusion we've reached, in mathematical language, is that metrical spacetime is a something.

24. Max Jammer, *Concepts of Space,* p. xvii.

Chapter 4